Knowledge index: source

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Complete structured-source inventory

All source cards are listed, including those not yet linked to a canonical synthesis. Downloaded literature remains in its original library folders.

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Scaling Laws for Fine-Grained Mixture of Experts	`source-doc`	0
Cost-Optimal Grouped-Query Attention for Long-Context Modeling	`source-doc`	0
Joint MoE Scaling Laws: Mixture of Experts Can Be Memory Efficient	`source-doc`	0
Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture-of-Experts Language Models	`source-doc`	0
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Scaling Data-Constrained Language Models	`source-doc`	0
To Repeat or Not To Repeat: Insights from Scaling LLM under Token-Crisis	`source-doc`	0
UniMax: Fairer and More Effective Language Sampling for Large-Scale Multilingual Pretraining	`source-doc`	0
Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset	`source-doc`	0
Larger Datasets Can Be Repeated More: A Theoretical Analysis of Multi-Epoch Scaling in Linear Regression	`source-doc`	0
Decouple Searching from Training: Scaling Data Mixing via Model Merging for Large Language Model Pre-training (DeMix)	`source-doc`	0
Olmix: A Framework for Data Mixing Throughout LM Development	`source-doc`	0
Prescriptive Scaling Laws for Data Constrained Training	`source-doc`	0
Scaling Laws for Mixture Pretraining Under Data Constraints	`source-doc`	0
GPT-4 Technical Report	`source-doc`	0
RULER: What's the Real Context Size of Your Long-Context Language Models?	`source-doc`	0
Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?	`source-doc`	0
Unveiling Downstream Performance Scaling of LLMs: A Clustering-Based Perspective (COD)	`source-doc`	0
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OLMo: Accelerating the Science of Language Models	source-doc	0
OLMoE: Open Mixture-of-Experts Language Models	source-doc	0
Phi-4 Technical Report	source-doc	0
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GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models	source-doc	0
Kimi K2: Open Agentic Intelligence	source-doc	0
Nemotron 3 Nano: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning	source-doc	0
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Qwen3 Technical Report	source-doc	0
Kimi K2.5: Visual Agentic Intelligence	source-doc	0
Kimi K3: Open Frontier Intelligence	source-doc	0
Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning	source-doc	0
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Model Soups: Averaging Weights of Multiple Fine-tuned Models	source-doc	0
Stop Wasting My Time! Saving Days of ImageNet and BERT Training with Latest Weight Averaging (LAWA)	source-doc	0
Tensor Programs VI: Feature Learning in Infinite-Depth Neural Networks	source-doc	0
Depthwise Hyperparameter Transfer in Residual Networks: Dynamics and Scaling Limit	source-doc	0
Early Weight Averaging meets High Learning Rates for LLM Pre-training	source-doc	0
DeepSeek LLM: Scaling Open-Source Language Models with Longtermism	source-doc	0
Scaling Optimal LR Across Token Horizons	source-doc	0
Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective	source-doc	0
Critical Batch Size Revisited: A Simple Empirical Approach to Large-Batch Language Model Training	source-doc	0

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Muon is Scalable for LLM Training	source-doc	0
Scaling and Transferability of Annealing Strategies in Large Language Model Training	source-doc	0
How to Allocate Your Tokens? Scaling Laws with Training Steps and Batch Size	source-doc	0
Small-Scale Experiments: Are We There Yet?	source-doc	0
Weight Decay Improves Language Model Plasticity	source-doc	0
Power Lines: Scaling Laws for Weight Decay and Batch Size in LLM Pre-training	source-doc	3
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Training Compute-Optimal Large Language Models (Chinchilla)	source-doc	0
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Language models scale reliably with over-training and on downstream tasks	source-doc	0
Resolving Discrepancies in Compute-Optimal Scaling of Language Models	source-doc	0
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Scaling Laws with Vocabulary: Larger Models Deserve Larger Vocabularies	source-doc	0
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Compare the mixture-swap scaling ladder for Monday's data swap	source-issue	4
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[post-training] Preserve final CruxEval transfer experiment	source-pr	0
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[datakit] Require core and Uncheatable evals in the decontam corpus	source-pr	0
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Source	Kind	Linked canonical pages
[identity sft] [sft] Verify HF exports against trained master weights	`source-pr`	0
[RL] Correctness (2/3): Fix reward and group-baseline rules	`source-pr`	0
[eval] Evaluate Hero at multiple context lengths	`source-pr`	0
Domain-Phase Data Mixture Swarm Experiments	`source-pr`	0
[rl] Add Delphi RL MATH-500 scaling probe launcher	`source-pr`	0
[evals] Add tagged MRCR context perplexity evaluation	`source-pr`	0
Add a general-purpose chat-SFT launcher on marin's ArtifactStep flow	`source-pr`	0
Generalize chat SFT into a shared sft_step + ModelSource (Grug 67B backend)	`source-pr`	3
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[datakit] Add a CoreWeave datakit MoE mixture swarm launcher	`source-pr`	0
[eval] Add RULER long-context eval to the evaluation launcher	`source-pr`	0
[datakit] emit global exact-dedup attributes	`source-pr`	0
datakit: coherent quality buckets from 88k GLM-5.2 labels and per-type calibration	`source-pr`	0
[datakit] Register the Harrier domain cluster assignments	`source-pr`	0
[datakit] Register the hero quality data as Datakit steps	`source-pr`	3
[datakit] Scale and validate decontamination	`source-pr`	0
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[datakit] Add transfer optimization for data mixtures	`source-pr`	0
Add long-context cooldown mixture	`source-pr`	0
Add 2x long-context treatment	`source-pr`	2
[evals] Run generative MRCR on Grug checkpoints	`source-pr`	0
Set 4x long-context skew	`source-pr`	0
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Add offline GRPO learner continuation	`source-pr`	0
[grug] Switch the hero mixture after checkpoint 108000	`source-pr`	4
Add GraphWalks evaluation to Evaldash	`source-pr`	0
Support context-length switches in the hero run	`source-pr`	0
[datakit] Score quality from the tokenize stage's token ids	`source-pr`	3
[eval] Run each evaluation as a StepSpec	`source-pr`	0

Jiaxuan Zou scientific scaling ladder synthesis

- Pretraining knowledge map and reading dependencies — `foundation-topic-map`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.

Marin Datakit released mixture experiment records: audited snapshot

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.

- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

Establishing Task Scaling Laws via Compute-Efficient Model Ladders (OLMo Task Ladder)

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

Predictable Scale: Part II, Farseer: A Refined Scaling Law in Large Language Models

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

KernelWiki: source-pinned knowledge engineering and validation practices

- Pretraining knowledge map and reading dependencies — `foundation-topic-map`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.

Local CPU record replays and analytical fixtures

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Knowledge coverage, unresolved inputs and review dates — `governance-coverage`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

Marin replication code audit: optimizer searches, MoE ladder and Datakit

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.
- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

Marin Data Pipeline Overview

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer (μ-Transfer)

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

Power Lines: Scaling Laws for Weight Decay and Batch Size in LLM Pre-training

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

Predictable Scale: Part I, Step Law

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

Current 96-H100 planning proposal

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Knowledge coverage, unresolved inputs and review dates — `governance-coverage`; inferred.

Agent MoE Experiment: 2x long-context skew at qk=1.75

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.

Test 4x long-context skew at qk=1.75

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.

Test 8x long-context skew at qk=1.75

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.
- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

Compare the mixture-swap scaling ladder for Monday's data swap

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

[post-training] RSI evaluation, audit and results

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.

Generalize chat SFT into a shared sft_step + ModelSource (Grug 67B backend)

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.

[datakit] Register the hero quality data as Datakit steps

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

Add 2x long-context treatment

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

[grug] Switch the hero mixture after checkpoint 108000

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

[datakit] Score quality from the tokenize stage’s token ids

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.